# A Ramanuj Patro

*AI & Full Stack Developer*

9556128887 ramanujpatro07@gmail.com LinkedIn GitHub Portfolio Berhampur, India

Computer Science undergraduate at SRM University passionate about AI, Full-Stack Development, and scalable software systems. Winner of India Innovates 2026 Hackathon, Oracle AI Foundations Certified, and Top 2 at Synapse Hackathon. Experienced in building AI-powered applications, backend systems, and modern web applications with Java, Python, React, Node.js, and Machine Learning.

## WORK EXPERIENCES

**Founder & Full Stack Developer** | Full-time — Feb 2026 - Present
*Rydin* — *Chennai*

- Built and launched a campus ride-sharing platform designed to connect students for shared transportation.
- Scaled the platform to **10,000+ active users** recorded on the platform.
- Developed the web application, authentication flow, and core user-facing functionality.
- Generated approximately **200K+ Instagram impressions/views** through product promotion and campus-focused outreach.

***Technologies / Skills Used :*** *React, Tailwind CSS*

## PROJECTS

### Synapse – AI Code Refactoring Assistant |

*Developer*

- Developed an **AI-powered code refactoring platform** that analyzes source code and generates optimized, cleaner, and more maintainable versions using Large Language Models.
- Implemented secure authentication, real-time code analysis, and an intuitive user interface for a seamless developer experience.
- Secured **2nd place** at the Confluence 1.0 Hackathon by delivering a functional AI-powered application under a competitive development timeline.

***Technologies / Tools Used :*** *React.js, Node.js, Express.js, MongoDB, Tailwind CSS*

### MDMARD — Multi-Domain Tactical Simulation | Ongoing Research |

*Research Project*

- Designed a modular simulation architecture incorporating physics, world state, observation, reward, and termination models using **Gymnasium**.
- Implemented a **PPO-based autonomous agent** with configurable dynamics, continuous/discrete actions, and reward shaping.
- Investigating extensions toward multi-agent coordination and multi-domain air-ground-sea simulation.

***Technologies / Tools Used :*** *Python, Gymnasium, PPO, NumPy, Pygame, Reinforcement Learning*

### Stelsion – AI-Powered Exoplanet Detection Platform |

*Developer*

- Developed a **1D CNN + Residual + Self-Attention model** for exoplanet detection from noisy **Kepler/TESS light curves**.
- Built using 1D Grad-CAM and attention visualization to analyze transit signals and false positives.

***Technologies / Tools Used :*** *Python, PyTorch, NumPy, SciPy, Scikit-learn, Kepler/TESS Data*

## SKILLS

| | |
|---|---|
| **Databases :** | MySQL, MongoDB |
| **Frameworks & Libraries :** | React.js, Tailwind CSS |
| **Languages :** | English, Hindi, Odia |
| **Programming Languages :** | Java, Python, C, C++, JavaScript, SQL |
| **Soft Skills :** | Communication, Leadership, Collaboration, Teamwork |
| **Tools & Platforms :** | Git, GitHub, VS Code, Postman |

## EDUCATIONS

**SRM Institute of Science and Technology** — 2025 - 2029
*B.Tech Computer Science* — *CGPA : 9*
Associate at E-Cell SRMIST; Technical and Business Member in Alexa Developers SRM.

## AWARDS & ACHIEVEMENTS

- India Innovates Hackathon Winner, Biggest Hackathon in Bharat Mandapam New Delhi - 2026
- Confluence Hackathon Runner-up - 2025
- IIT Bombay Techfest Hackathon Finalist - 2025

## CERTIFICATIONS

- **Oracle Cloud Infrastructure 2025 — AI Foundations Associate**, Oracle
- **DSA with Java**, Apna College